Reader: Do we really have to resort to a similar line of reasoning, based on the definition of the limit of function at a point, each time we have to find the limit of $\left(\frac{f(x)}{g(x)}\right)$ when both $\lim_{x \to 0} f(x) = 0$ and $\lim_{x \to 0} g(x) = 0$?

Author: No, of course not. The situation we are speaking about is known as an $\boxed{\text{indeterminate form}}$ of the type $\frac{0}{0}$. There are rules which enable one to analyze such a situation in a relatively straightforward manner, and, so to say, "resolve the indeterminacy". In practice it is usually not difficult to solve the problem of existence of the limit of a function $\left(\frac{f(x)}{g(x)}\right)$ at a specific point and finds its value (if it exists). A few rules of evaluation of indeterminate forms of the type $\frac{0}{0}$ (and other types as well) will be discussed later. A systematic analysis of such rules, however, goes beyond the scope of our dialogues.

It is important to stress here the following principle (which is significant for further considerations): ~~although the~~ theorem on the limit of the ratio is not valid in the cases when $\boxed{\lim_{x \to 0} g(x) = 0}$, the limit of a function $\left(\frac{f(x)}{g(x)}\right)$ at a point a may exist if $\lim_{x \to 0} f(x) = 0$. The example of the limit of the function $\frac{\sin x}{x}$ at $x = 0$ is a convincing illustration of this principle.

Reader: Presumably, a similar situation make take place for numerical sequences as well?

Author: It certainly may. Here is a simple example:

$$(x_n) = 1, \frac{1}{8}, \frac{1}{27}, \frac{1}{64}, \ldots, \frac{1}{n^3}, \ldots$$

$$\left(\lim_{n \to \infty} x_n = 0\right)$$

$$(y_n) = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots, \frac{1}{n}, \ldots$$

$$\left(\lim_{n \to \infty} y_n = 0\right)$$

It is readily apparent that the limit of the sequence $\left(\frac{x_n}{y_n}\right)$ is the limit of the sequence $\left(\frac{1}{n^2}\right)$. This limit does exist and is equal to zero.